

QUESTION 1

- (a) Solve $2x^2 > 2 - 3x$.
- (b) Differentiate $\frac{1}{4 - x^2}$
- (c) Find the exact value of $\int_0^{\frac{\pi}{4}} \cos x \sin^2 x \, dx$.
- (d) A committee of 5 is to be chosen from 6 boys and 3 girls. Find the probability that the committee contains a particular boy X and a particular girl Y.
- (e) Find the acute angle between the lines $y = 3x - 2$ and $x - 2y = 5$

Question 2 (START A NEW PAGE)

- (a) (i) Draw a sketch of $y = 2\sin^{-1}x$. State the domain and range.
- (ii) A region R is bounded by the curve $y = 2\sin^{-1}x$, the x-axis and the line $x = 1$.

Find the exact area of the region R.

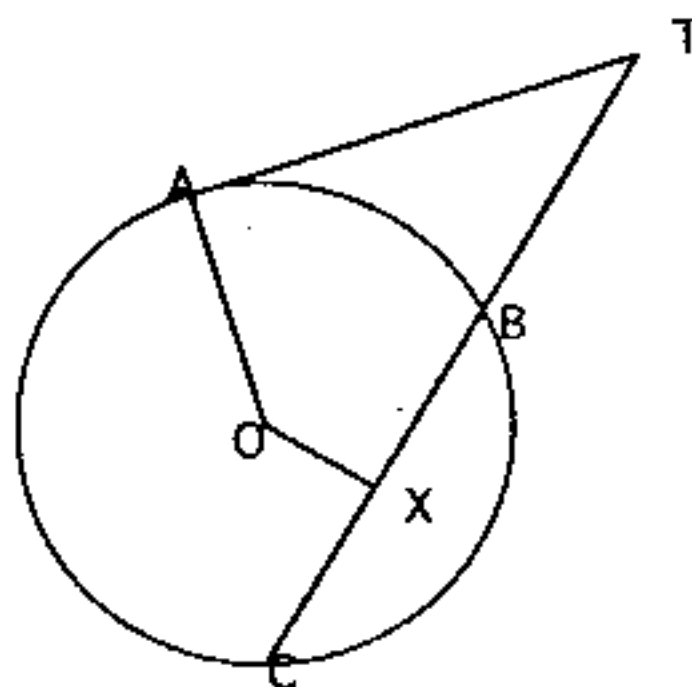
- (b) Find $\int_0^1 x^2(x^3 + 1)^4 \, dx$ using the substitution $u = x^3 + 1$

QUESTION 3 (START A NEW PAGE)

- (a) A point moves along the curve $y = \ln x^2$. The x co-ordinate of the point changes at the rate of 4 units per second. At what rate is the y co-ordinate increasing when $x = 3$?
- (b) Find the term independent of x in $(x - \frac{1}{x})^6$.
- (c) When A and B play chess, the probability of either winning a game is always $\frac{1}{4}$ and the probability of the game being drawn is always $\frac{1}{2}$. Find the probability of A winning at least four games out of five. (Answer correct to four decimal places)
- (d) Find the general solution to the equation $\tan \theta = \sin 2\theta$

QUESTION 4 (START A NEW PAGE)

- (a) A, B, C are three points on a circle, centre O. The tangent at A meets CB produced at T. X is the mid point of BC. Prove that:-
- AOXT is a cyclic quadrilateral
 - $\angle AOT = \angle AXT$.



- (b) $S_n = a + ar + ar^2 + \dots + ar^{n-1}$.

- Using mathematical induction or otherwise, prove that $S_n = \frac{a(1 - r^n)}{1 - r}$
- Write down an expression for the limiting sum of a G.P.
- State the values of r for which it exists.
- If θ is not a multiple of $\frac{\pi}{2}$ and if p and q are given as sums of the following infinite geometric series:-

$$p = 1 + \cos^2\theta + \cos^4\theta + \dots$$

$$q = 1 + \sin^2\theta + \sin^4\theta + \dots$$

prove that $p + q = pq$.

QUESTION 5 (START A NEW PAGE)

- (a) A projectile travels in a parabolic path. The angle of projection is 60° and the velocity at which it is projected is 500 m/sec.
- Derive the equations of motion for the projectile in flight. (Air resistance is to be neglected and the acceleration due to gravity is to be taken as 10ms^{-2})
 - Find the range
 - Find the greatest height reached.
- (b) P $(2ap, ap^2)$ and Q $(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.
- Write down the equations of the normals at P and Q.
 - Find the co-ordinates of R, the point of intersection of the normals, in terms of p and q .
 - If $pq = -2$ find the cartesian equation of the locus of R.

QUESTION 6 (START A NEW PAGE)

- (a) A particle moving with simple harmonic motion makes 100 complete oscillations per minute. Its maximum speed is 10m per sec. Find:-
 (i) the exact period of the motion
 (ii) the amplitude of the motion.
 (iii) the time taken to move from the centre of the oscillation to a point which is distant two-thirds of the amplitude from the centre (Give answer in secs to two significant figures)
- (b) (i) Sketch the curve $y = 2\cos x - 1$ for $-\pi \leq x \leq \pi$. Mark clearly where the graph crosses each axis.
 (ii) Find the volume generated by the rotation through a complete revolution about the x-axis of the region between the x-axis and that part of the curve $y = 2\cos x - 1$ for which $|x| \leq \pi$ and $y \geq 0$.

Question 7 (START A NEW PAGE)

- (a) The constant acceleration of a train is 1metre per sec per sec and its constant retardation is 3 metres per sec per sec.
 i) Sketch a velocity-time graph assuming the train starts from rest accelerates in a straight line and immediately decelerates in a straight line until it is again at rest.
 ii) Find the time taken for a journey of 1Km. given the journey described in (i)
- (b) The integers a,b,d are connected by the relation $a = b + d$.
 (i) Use the binomial expansion of $(b + d)^n$, where n is a positive integer, to show that $a^n - b^{n-1} (b + nd)$ is divisible by d^2 .
 (ii) In the result of part (i) replace b by a - d. Hence show that if a is the first term, d the common difference and l the nth term of an arithmetic progression , then $a^n - l (a - d)^{n-1}$ is divisible by d^2 .
 (iii) Deduce that $5^{682} - 2^{692}$ is divisible by 9.

THIS IS THE END OF THE PAPER

$$x < -2, x > \frac{1}{2}$$

$$(b) \frac{2x}{(4-x^2)^2}$$

$$(c) \frac{1}{6\sqrt{2}}$$

$$(d) \frac{\binom{7}{3}}{\binom{9}{5}}$$

$$(e) 45^\circ$$

$$Q2 (a) (i) D: -1 \leq x \leq 1$$

$$R: -\pi \leq y \leq \pi$$

$$(ii) \pi - 2$$

$$(b) \frac{31}{15}$$

$$Q3 (a) \frac{8}{3} \text{ v/s}$$

$$(b) C_u = -20$$

$$(c) 0.0156$$

$$(d) \theta = n\pi \pm \pi/4$$

$$Q4 (a) -$$

$$(b) (i) -$$

$$(ii) \frac{a}{1-r}$$

$$(iii) |x| < 1$$

$$(iv) -$$

$$Q5 (a) -$$

$$(ii) 12500\sqrt{3} \text{ m}$$

$$(iii) 9375 \text{ m}$$

$$(b) (i) -$$

$$(ii) (-apq(p+q), c(p^2+pq+q^2+2))$$

$$(iii) x^2 = 4a(y-4a)$$

$$Q6 (a) (i) 0.6 \text{ s}$$

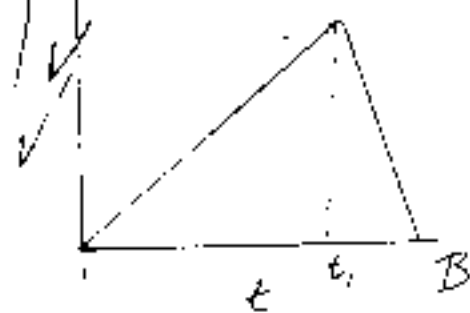
$$(ii) a = \frac{3}{a} \text{ m}$$

$$(iii) 0.070 \text{ rad}$$

$$(b) (i) -$$

$$ii -3\sqrt{3}a + 2a^2$$

$$Q7 (a) (i) \downarrow$$



$$(ii) \frac{2\sqrt{3}}{3} \text{ m}$$

$$(b) -$$